

Technical Document #65: Computer Vision - Comprehensive Overview

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Image generation creates new images from random noise or text descriptions. Diffusion models like Stable Diffusion produce photorealistic images.

Face recognition identifies or verifies faces in images. DeepFace and FaceNet achieve near-human accuracy on face verification tasks.

Semantic segmentation classifies each pixel in an image into a category. It is used in autonomous driving and medical image analysis.

Visual question answering (VQA) answers questions about images. It requires both visual understanding and language comprehension.

Image classification assigns a label to an image from a fixed set of categories. CNNs like ResNet and EfficientNet achieve superhuman performance on benchmark datasets.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Face recognition identifies or verifies faces in images.
- Video understanding analyzes temporal dynamics in video sequences.
- Semantic segmentation classifies each pixel in an image into a category.